

資 1 | A. 宋辰星 01257001

1. (a)

$$\begin{aligned}C &\equiv 11b \pmod{13} \\&\equiv 99 \pmod{13} \\&\equiv 8 \pmod{13}\end{aligned}$$

$$\begin{aligned}b &\equiv 9 \pmod{13} \\11b &\equiv 99 \pmod{13}\end{aligned}$$

$$A: C = 8 \quad (0 \leq C \leq 12)$$

(b)

$$\begin{aligned}C &\equiv 2a + 3b \pmod{13} \\&\equiv 35 \pmod{13} \\&\equiv 9 \pmod{13}\end{aligned}$$

$$\begin{aligned}2a &\equiv 8 \pmod{13} \\3b &\equiv 27 \pmod{13}\end{aligned}$$

$$A: C = 9 \quad (0 \leq C \leq 12)$$

(c)

$$\begin{aligned}C &\equiv a^3 - b^3 \pmod{13} \\&\equiv 64 - 729 \pmod{13} \\&\equiv -665 \pmod{13} \\&\equiv -2 \pmod{13} \\&\equiv 11 \pmod{13}\end{aligned}$$

$$\begin{aligned}a^3 &\equiv 4^3 \pmod{13} \\b^3 &\equiv 9^3 \pmod{13}\end{aligned}$$

$$\begin{array}{r}57 \\13 \overline{) 665} \\ \underline{65} \\ 15 \\ \underline{13} \\ 2\end{array}$$

$$A: C = 11 \quad (0 \leq C \leq 12)$$

2.

$$(a) (177 \bmod 31 \cdot 270 \bmod 31) \bmod 31$$

$$\begin{array}{r} 5 \\ 31 \overline{) 177} \\ \underline{155} \\ 22 \end{array}$$

$$177 \bmod 31 = 22$$

$$270 \bmod 31 = 22$$

$$\Rightarrow 44 \bmod 31$$

$$= 13$$

$$(b) (-133 \bmod 23 + 261 \bmod 23) \bmod 23$$

$$= (-133 + 261) \bmod 23$$

$$= 128 \bmod 23$$

$$= 13$$

$$\begin{array}{r} 5 \\ 23 \overline{) 128} \\ \underline{115} \\ 13 \end{array}$$

$$(c) (32^3 \bmod 13)^2 \bmod 11$$

$$= ((32 \bmod 13)^3 \bmod 13)^2 \bmod 11$$

$$= (6^3 \bmod 13)^2 \bmod 11$$

$$= (216 \bmod 13)^2 \bmod 11$$

$$= 64 \bmod 11$$

$$= 9$$

$$\begin{array}{r} 16 \\ 13 \overline{) 216} \\ \underline{130} \\ 86 \\ \underline{78} \\ 8 \end{array}$$

$$(d) (99^2 \bmod 32)^3 \bmod 15.$$

$$= ((99 \bmod 32)^2 \bmod 32)^3 \bmod 15.$$

$$= (3^2 \bmod 32)^3 \bmod 15$$

$$= 729 \bmod 15$$

$$= 9$$

$$\begin{array}{r} 32 \overline{) 99} \\ \underline{96} \\ 3 \end{array}$$

3.

$$(a) (97644)_{10} = (1011110101100)_{2}$$

$$\begin{array}{r} 2 \overline{) 97644} \\ \underline{48822} \quad -0 \\ 2 \overline{) 24411} \quad -0 \\ 2 \overline{) 12205} \quad -1 \\ 2 \overline{) 6102} \quad -1 \\ 2 \overline{) 3051} \quad -0 \\ 2 \overline{) 1525} \quad -1 \end{array}$$

$$(b) (1010110101)_{2} = (1 \cdot 2^9 + 0 \cdot 2^8 + 1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0)$$

$$= (512 + 128 + 32 + 16 + 4 + 1)$$

$$= (693)_{10}.$$

$$\begin{array}{r} 2 \overline{) 762} \quad -1 \\ 2 \overline{) 381} \quad -0 \\ 2 \overline{) 190} \quad -1 \end{array}$$

$$(c) (423)_8 = (100010011)_2.$$

$$(d) (101010101010)_2 = (5252)_8.$$

$$\begin{array}{r} 2 \overline{) 95} \quad -0 \\ 2 \overline{) 47} \quad -1 \\ 2 \overline{) 23} \quad -1 \\ 2 \overline{) 11} \quad -1 \\ 2 \overline{) 5} \quad -1 \\ 2 \overline{) 2} \quad -1 \\ 1 \quad -0 \end{array}$$

$$(e) (135AB)_{16} = (0001010101010101)_{2}$$

$$= (232653)_{8}.$$

$$(f) (BADFACED)_{16}$$

$$= (10111010110111010100110101)_{2}$$

$$= (27267726355)_{8}.$$

$$(g) (10110111011)_{2} = (5573)_{8}.$$

4. (a)

$$\begin{array}{r} 64 \\ 32 \\ 16 \\ 8 \\ \hline 119 \end{array}$$

$$(1000111)_2, (1110111)_2$$

$$\left\{ \begin{array}{l} \text{sum: } (10111110)_2 \\ \text{product: } (10000100000001) \end{array} \right.$$

$$(2^6 + 2^2 + 2 + 1)_{10} \quad (2^6 + 2^5 + 2^4 + 2^2 + 2 + 1)$$

$$(64 + 4 + 3)_{10} \quad (64 + 32 + 16 + 4 + 3)$$

$$71_{10}$$

$$119_{10}$$

$$\begin{array}{r} 119 \\ \times 71 \\ \hline 833 \\ 8449 \end{array}$$

$$\begin{array}{l} 8449 \\ = 8192 = (2^{10}) \times 2^3 \\ + 257 \\ + 256 = 2^8 \\ + 1 = 2^0 \end{array}$$

(b)

$$(112)_3, (210)_3$$

$$\left\{ \begin{array}{l} \text{sum: } (1022)_3 \\ \text{product: } (101220)_3 \end{array} \right.$$

$$3^2 + 3 + 2 \quad 2 \cdot 3 + 10$$

$$\begin{array}{r} 3 \overline{) 11}^{-2} \quad 9 + 5 = 14 \quad 18 + 3 = 21 \\ 3 \overline{) 13}^{-2} \end{array}$$

$$1^{-0} \quad 35_{10} = 1022$$

$$294_{10} = 101220$$

$$\begin{array}{r} 14 \\ + 21 \\ \hline 14 \\ + 28 \\ \hline 214 \end{array}$$

$$\begin{array}{r} 3 \overline{) 294} \\ 3 \overline{) 11}^{-0} \\ 3 \overline{) 12}^{-2} \\ 3 \overline{) 10}^{-2} \\ 3 \overline{) 13}^{-1} \\ 1^{-0} \end{array}$$

4. (c)

$$(763)_8, (147)_8$$

$$\begin{array}{r} 11 \\ 763 \\ + 147 \\ \hline 1132 \end{array}$$

$$\begin{cases} \text{sum: } (1132)_8 \\ \text{product: } (144305)_8 \end{cases}$$

$$\begin{array}{r} 52 \quad 21 \quad 35 \\ 763 \quad 44 \quad 35 \\ \times 147 \quad 54 \quad 31 \\ \hline 276145. \\ 3714 \\ 763 \\ \hline 144305 \end{array}$$

(d)

$$(1AE)_{16}, (BBC)_{16}$$

$$\begin{array}{r} 11 \\ 1AE \\ + BBC \\ \hline D6A \end{array}$$

$$\begin{cases} \text{sum: } (D6A)_{16} \\ \text{product: } (13B5C8)_{16} \end{cases}$$

$$\begin{array}{r} 8A \\ 1AE \\ \times BBC \\ \hline 17928. \\ 127A \\ 127A \\ \hline 13B5C8. \end{array}$$

$$\begin{array}{r} 27. \\ 22 \\ 20 \\ 12 \\ \times 14 \\ \hline 48 \\ 12 \\ 68 \\ 76 \\ 8 \\ \hline 154 \end{array}$$

110.
(119)

5.

$$(a) \ 11^{644} \bmod 645$$

$$\begin{array}{r} 645 \\ \times 12 \\ \hline 133 \\ \times 5 \\ \hline 645 \\ \hline 5 \end{array} \quad 644 = 512 + 128 + 4$$

$$= 2^9 + 2^7 + 2^2$$

$$\Rightarrow (645)_{10} = (1010000100)_2$$

$$11^{644} = 11^{2^9 + 2^7 + 2^2}$$

$$11^{2^0} \bmod 645 = 11$$

$$11^{2^1} \bmod 645 = 121$$

$$\checkmark 11^{2^2} \bmod 645 = 121^2 \bmod 645 = 451$$

$$11^{2^3} \bmod 645 = 451^2 \bmod 645 = 226$$

$$11^{2^4} \bmod 645 = 226^2 \bmod 645 = 121$$

$$11^{2^5} \bmod 645 = 121^2 \bmod 645 = 451$$

$$11^{2^6} \bmod 645 = 226$$

$$\checkmark 11^{2^7} \bmod 645 = 121$$

$$11^{2^8} \bmod 645 = 451$$

$$\checkmark 11^{2^9} \bmod 645 = 226$$

$$11^{644} \bmod 645 = (11^{2^2} \cdot 11^{2^7} \cdot 11^{2^9}) \bmod 645$$

$$= (451 \times 121 \times 226) \bmod 645$$

$$= 1$$

5. (b)

$$3^{2003} \bmod 99.$$

$$2003 = (\underbrace{11111010011}_{{10\ 9\ 8\ 7\ 6\ 5\ 4\ 3\ 2\ 1\ 0}})_2$$

$$3^{2003} = 3^{2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^4 + 2^1 + 2^0}$$

$$\begin{array}{r} 2 \overline{) 2003} \\ 2 \overline{) 1001} \rightarrow 1 \\ 2 \overline{) 500} \rightarrow 1 \\ 2 \overline{) 250} \rightarrow 0 \\ 2 \overline{) 125} \rightarrow 0 \\ 2 \overline{) 62} \rightarrow 1 \\ 2 \overline{) 31} \rightarrow 1 \\ 2 \overline{) 15} \rightarrow 1 \\ 2 \overline{) 7} \rightarrow 1 \\ 2 \overline{) 3} \rightarrow 1 \\ 1 \rightarrow 1 \end{array}$$

$$(3^{2^0}) \bmod 99 = 3.$$

$$(3^{2^1}) \bmod 99 = 9 \bmod 99 = 9.$$

$$3^{2^2} \bmod 99 = 81 \bmod 99 = 81$$

$$3^{2^3} \bmod 99 = 81^2 \bmod 99 = 27.$$

$$(3^{2^4}) \bmod 99 = 27^2 \bmod 99 = 36.$$

$$3^{2^5} \bmod 99 = 36^2 \bmod 99 = 9.$$

$$(3^{2^6}) \bmod 99 = 81$$

$$(3^{2^7}) \bmod 99 = 27$$

$$(3^{2^8}) \bmod 99 = 36.$$

$$(3^{2^9}) \bmod 99 = 9$$

$$(3^{2^{10}}) \bmod 99 = 81$$

$$\begin{aligned} 3^{2003} \bmod 99 &= (3 \times 9 \times \boxed{27} \times 36 \times 81 \times 27 \times 3 \times 9 \times 81) \bmod 99 \\ &= 27 \end{aligned}$$

5 (c)

$$123^{1001} = 123 \cdot (123^{100})^{10}$$

$$\because 123^{100} \equiv 1 \pmod{101} \text{ by Fermat's little Thm,}$$

$$\begin{aligned}\therefore 123^{1001} \pmod{101} &= 123(123^{100})^{10} \pmod{101} \\ &= (123 \pmod{101})(123^{100})^{10} \pmod{101} \\ &= 22 \times 1 \pmod{101} \\ &= 22_{\times}\end{aligned}$$

(d)

$$7^{121} \pmod{13}$$

$$7^{12} \equiv 1 \pmod{13}$$

$$7^{121} = (7^{12})^{10} \times 7$$

$$\Rightarrow 7^{121} \pmod{13}$$

$$= (7^{12})^{10} \times 7 \pmod{13}$$

$$= [(7^{12})^{10} \pmod{13} \times (7 \pmod{13})] \pmod{13}$$

$$= (1 \times 7) \pmod{13}$$

$$= 7_{\times}$$

5 (e)

$$23^{40} \equiv 1 \pmod{41}$$

$$23^{1002} = (23^{40})^{25} \times 23^2$$

$$\Rightarrow 23^{1002} \pmod{41}$$

$$= [(23^{40})^{25} \times 23^2] \pmod{41}$$

$$= [(23^{40})^{25} \pmod{41} \times 23^2 \pmod{41}] \pmod{41}$$

$$= (1 \times 37) \pmod{41}$$

$$= 37$$

(f) $\gcd(1529, 14039)$

$$\begin{array}{r} 278 \\ 1529 \\ \times 9 \\ \hline 13761 \end{array}$$

$$\begin{array}{r} 3910 \\ 14039 \\ 13761 \\ \hline 278 \\ \times 5 \\ \hline 1390 \end{array}$$

$$\begin{array}{r} 410 \\ 1529 \\ 1390 \\ \hline 139 \end{array}$$

$$14039 = 1529 \times 9 + 278$$

$$1529 = 278 \times 5 + 139$$

$$278 = 139 \times 2 + 0$$

$$\Rightarrow \gcd(1529, 14039) = 139$$

(g) $\gcd(1111, 0)$

$\Rightarrow$ 未定義

6.

(9)

$$\gcd(117, 213).$$

$$\begin{array}{r} 213 \overset{10}{=} \\ 117 \\ \hline 96 \end{array}$$

$$\begin{array}{r} 117 \\ 96 \\ \hline 21 \end{array}$$

$$\textcircled{213} = \textcircled{117} \times 1 + 96.$$

$$117 = 96 \times 1 + 21$$

$$96 = 21 \times 4 + 12$$

$$21 = 12 \times 1 + 9.$$

$$12 = 9 \times 1 + 3.$$

$$9 = \textcircled{3} \times 3 + 0.$$

$$> 13 \quad 3 = 12 - 9.$$

$$= 1 \times 12 + (-1) \times 9$$

$$= 1 \times 12 + (-1) \times (21 - 12)$$

$$= 12 \times 12 + 21 \times (-1)$$

$$= 2(96 - 4 \cdot 21) - 21$$

$$= 2 \cdot 96 - 9 \cdot 21$$

$$= 2 \cdot 96 - 9(117 - 96)$$

$$= 11 \cdot 96 - 9 \cdot 117$$

$$= 11 \cdot (213 - 117) - 9 \cdot 117$$

$$= \underline{11 \cdot 213 - 26 \cdot 117}$$

6. (b)

$$\gcd(124, 323)$$

$$323 = 124 \times 2 + 75$$

$$124 = 75 \times 1 + 49$$

$$75 = 49 \times 1 + 26$$

$$49 = 26 \times 1 + 23$$

$$26 = 23 \times 1 + 3$$

$$23 = 3 \times 7 + 2$$

$$3 = 2 \times 1 + 1$$

$$2 = 1 \times 2$$

$$\gcd(124, 323) = 1$$

$$\begin{array}{r} 10 \quad 10 \\ 2 \overline{) 323} \\ \underline{- 248} \\ 675 \\ \underline{- 49} \\ 26 \end{array} \quad \begin{array}{r} 10 \quad 10 \\ 124 \overline{) 323} \\ \underline{- 75} \\ 49 \\ \underline{- 26} \\ 23 \end{array}$$

$$\begin{array}{r} 17 \\ 26 \overline{) 43} \end{array}$$

$$1 = 3 - 2$$

$$= 3 - (23 - 3 \times 7)$$

$$= 3 \times 8 - 23$$

$$= 8(26 - 23) - 23$$

$$= 8 \cdot 26 - 9 \cdot 23$$

$$= 8 \cdot 26 - 9(49 - 26)$$

$$= 17 \times 26 - 9 \cdot 49$$

$$= 17(75 - 49) - 9 \cdot 49$$

$$= 17 \cdot 75 - 26 \cdot 49$$

$$= 17(75) - 26(124 - 75)$$

$$= 43 \cdot 75 - 26 \cdot 124$$

$$= 43(323 - 124 \times 2) - 26 \cdot 124$$

$$= 43 \cdot 323 - 112 \cdot 124$$

7. (a)

$$4x \equiv 5 \pmod{9}$$

$$9 = 4 \times 2 + 1$$

$$4 = 1 \times 4 + 0.$$

$$1 = 9 - 4 \times 2$$

$$= 9 \times 1 + (-2) \times 4.$$

$$\text{inverse} = -2 + k,$$

$$17 \times 4x \equiv 17 \times 5 \pmod{9}$$

$$x \equiv 35 \pmod{9}$$

$$x \equiv 8 \pmod{9}$$

$$\begin{array}{r} 2 \times 5 \\ 2 \times 5 \\ \hline 8 \end{array}$$

(b)

$$34x \equiv 77 \pmod{89}$$

$$\begin{array}{r} 89 \\ 68 \\ \hline 21 \end{array}$$

$$89 = 34 \times 2 + 21$$

$$34 = 21 \times 1 + 13$$

$$21 = 13 \times 1 + 8.$$

$$13 = 8 \times 1 + 5$$

$$8 = 5 \times 1 + 3.$$

$$5 = 3 \times 1 + 2.$$

$$3 = 2 \times 1 + 1$$

$$2 = 1 \times 2 + 0.$$

$$\text{gcd} = 1$$

7. b) 續:

$$1 = 3 - 2.$$

$$= 3 - (5 - 3)$$

$$= 2 \times 3 - 5$$

$$= 2(8 - 5) - 5$$

$$= 2 \times 8 - 3 \times 5$$

$$= 2 \times 8 - 3(13 - 8)$$

$$= 5 \times 8 - 3 \times 13$$

$$= 5(21 - 13) - 3 \times 13.$$

$$= 5 \times 21 - 8 \times 13.$$

$$= 5 \times 21 - 8(34 - 21)$$

$$= 13 \times 21 - 8 \times 34$$

$$= 13(89 - 34 \times 2) - 8 \times 34$$

$$= 13 \times 89 - \textcircled{34} \times 34$$

$$= 6 \times 89 - 34$$

$$\text{Inverse} : -34 + K \Rightarrow 55.$$

$$55 \times 34 \times X \equiv 55 \cdot 177 \pmod{89}$$

$$X \equiv 4235 \pmod{89}$$

$$X \equiv 52 \pmod{89}$$

$$4235 \pmod{89} \\ = 52.$$

7. (c)

$$15x^2 + 19x \equiv 5 \pmod{11}$$

$$15x^2 + 19x + 6 \equiv 0 \pmod{11}$$

$$\begin{array}{cc} 3 & 2 \\ 5 & 3 \end{array}$$

$$(3x+2)(5x+3) \equiv 0 \pmod{11}$$

①:

$$3x+2 \equiv 0 \pmod{11}$$

$$3x + \cancel{2} + 9 \equiv 9 \pmod{11}$$

$$3x \equiv 9 \pmod{11}$$

②:

$$(5x+3) \equiv 0 \pmod{11}$$

$$5x + \cancel{3} + 8 \equiv 8 \pmod{11}$$

$$5x \equiv 8 \pmod{11}$$

$$11 = 2 \times 5 + 1$$

$$5 = 1 \times 5 + 0$$

$$1 = 11 - 2 \times 5 \quad \text{inverse} = -2 + k \Rightarrow 9$$

$$9 \times 5x \equiv 9 \times 8 \pmod{11}$$

$$x \equiv 12 \pmod{11}$$

$$\begin{array}{r} 6 \times 2 \\ 12 \\ 66 \\ \hline 6 \end{array}$$

$$x \equiv 6 \pmod{11}$$

$$11 = 3 \times 3 + 2$$

$$3 = 2 \times 1 + 1$$

$$1 = 1 \times 1 + 0$$

$$1 = 3 - 2$$

$$= 3 - (11 - 3 \times 3)$$

$$= 4 \times 3 - 11$$

$$4 \times 3x \equiv 4 \times 9 \pmod{11}$$

$$x \equiv 36 \pmod{11}$$

$$x \equiv 3 \pmod{11}$$

$$A: x = \{3, 6\}$$

7. (d)

$$x \equiv 5 \pmod{6},$$

$$6 = 2 \times 3$$

$$x \equiv 3 \pmod{10},$$

$$10 = 2 \times 5$$

$$x \equiv 8 \pmod{15},$$

$$15 = 3 \times 5.$$

$$x \equiv 5 \pmod{6}$$

$$\Rightarrow \begin{cases} x \equiv 5 \equiv 1 \pmod{2} \checkmark \\ x \equiv 5 \equiv 2 \pmod{3} \checkmark \end{cases}$$

$$x \equiv 3 \pmod{10}$$

$$\Rightarrow \begin{cases} x \equiv 3 \equiv 1 \pmod{2} \checkmark \\ x \equiv 3 \pmod{5} \checkmark \end{cases}$$

$$x \equiv 8 \pmod{15}$$

$$\Rightarrow \begin{cases} x \equiv 8 \equiv 2 \pmod{3} \checkmark \\ x \equiv 8 \equiv 3 \pmod{5} \checkmark \end{cases}$$

$$x \equiv 1 \pmod{2}$$

$$\Rightarrow x = 2t + 1$$

$$2t + 1 \equiv 2 \pmod{3}$$

$$2t \equiv 1 \pmod{3}$$

$$t \equiv 2 \pmod{3}$$

$$\Rightarrow t = 3s + 2$$

$$\Rightarrow x = 2(3s + 2) + 1$$

$$\Rightarrow x = 6s + 5$$

$$x \equiv 3 \pmod{5}$$

$$6s + 5 \equiv 3 \pmod{5}$$

$$s \equiv 3 - (5s + 5) \pmod{5}$$

$$s \equiv 3 \pmod{5}$$

$$\Rightarrow s = 5k + 3$$

$$x = 6s + 5$$

$$= 6(5k + 3) + 5, 30k + 23.$$

$$\Rightarrow x = 30k + 23$$

8. (a)

$$P(n) : 1 \times 2 + 2 \times 3 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

$$n=1 :$$

$$1 \cdot 2 = 2$$

$$\frac{1(1+1)(2+1)}{3} = \frac{1 \times 2 \times 3}{3} = 2$$

let $P(k)$ is true :

$$1 \times 2 + 2 \times 3 + \dots + k(k+1) = \frac{k(k+1)(k+2)}{3}$$

$$P(k+1) :$$

$$1 \times 2 + 2 \times 3 + \dots + k(k+1) + (k+1)(k+2)$$

$$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$= \left(\frac{k}{3} + 1\right)(k+1)(k+2)$$

$$= \left(\frac{k+3}{3}\right)(k+1)(k+2)$$

$$= \frac{(k+1)(k+2)(k+3)}{3}$$

$$= \frac{(k+1)[(k+1)+1][(k+1)+2]}{3}$$

$\Rightarrow P(k+1)$ also true. $\Rightarrow$ 得證.

8. b,

$$\text{在 } [a^{k+1} = \frac{a^k \cdot a^k}{a^{k-1}} = \frac{1 \cdot 1}{1} = 1] \text{ 中,}$$

條件需為 $a^k = a^{k+1} = 1$, 所以,

$a^1 = 1$ 也要在 Basis step 中證明。